GATE ME 90 Days Challenge

Day 68/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
General Aptitude	GA Revision	32 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on a full revision of General Aptitude: verbal ability (tenses, antonyms, sentence completion, critical reasoning), quantitative aptitude (profit and loss, work, ages, mensuration, probability), analytical reasoning (directions, number series, syllogisms), data interpretation from a line graph and spatial aptitude (paper folding and shortest paths on a surface). Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

Select the option that correctly fills the blank: By the time the inspector arrives tomorrow, we _____ the pressure test.

(A) will finish
(B) will have finished
(C) have finished
(D) had finished

Q2 MCQ

Select the word that is most nearly OPPOSITE in meaning to FRUGAL.

(A) thrifty
(B) careful
(C) extravagant
(D) modest

Q3 MCQ

Select the most appropriate word to fill the blank: The trial results were _____: they neither clearly supported nor clearly ruled out the new design.

(A) inconclusive
(B) conclusive
(C) impressive
(D) exhaustive

Q4 NAT

A shopkeeper sells an article for ₹1140 and suffers a loss of 5 %. The selling price (in ₹) at which he must sell it to gain 10 % is _____

Q5 NAT

A and B together can finish a job in 12 days, B and C together in 15 days, and C and A together in 20 days. The number of days A alone would take to finish the job is _____

Q6 MCQ

The ratio of the present ages of a father and his son is 7 : 2. After 10 years the ratio will be 9 : 4. The present age of the father is

(A) 30 years
(B) 40 years
(C) 42 years
(D) 35 years

Q7 MCQ

Ravi walks 4 km north, turns right and walks 3 km, turns right again and walks 8 km, and then turns left and walks 3 km. How far is he from the starting point, and in which direction?

(A) 7.21 km, south-east
(B) 5 km, north-east
(C) 7.21 km, south-west
(D) 4 km, south

Q8 NAT

The next term of the series 4, 9, 25, 49, 121, ___ is _____

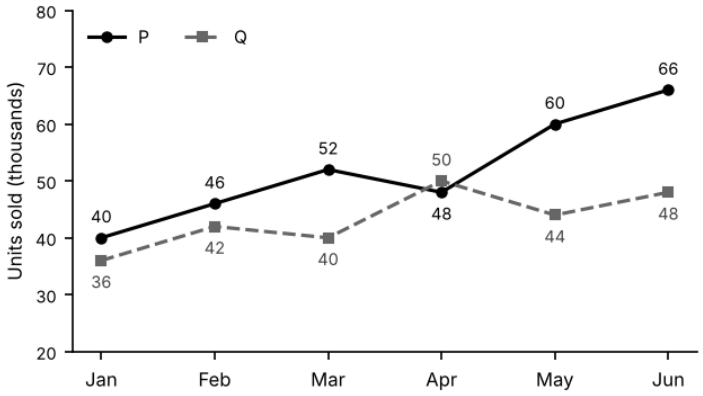
Q9 MCQ

Statements: Some valves are pipes. No pipe is a pump. Conclusions: I. Some valves are not pumps. II. Some pumps are valves. Which conclusion(s) follow?

(A) Only II follows
(B) Only I follows
(C) Both I and II follow
(D) Neither I nor II follows

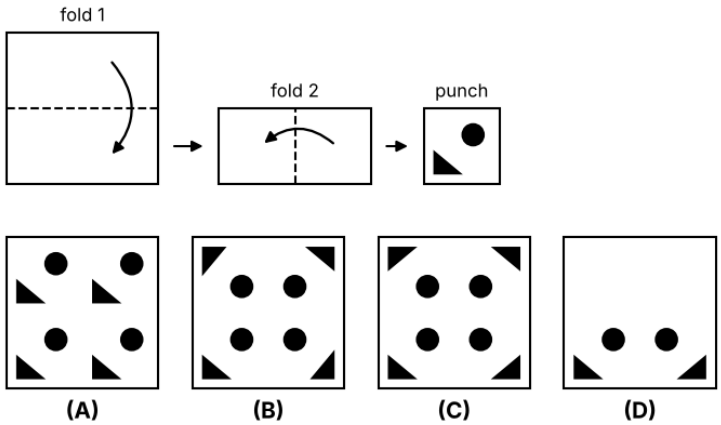
Q10 NAT

The line graph shows the monthly sales of two products P and Q from January to June. By what percentage does the total six-month sales of P exceed that of Q? _____



Q11 MCQ

A square sheet of paper is folded twice (first the top half down onto the bottom half, then the right half onto the left half) and two holes are punched through all the layers, as shown. Which option shows the sheet when it is unfolded?



(A) Figure (A)
(B) Figure (B)
(C) Figure (C)
(D) Figure (D)

Q12 NAT

An ant crawls on the curved surface of a closed cylindrical can of radius 7 cm and height 30 cm, from a point on the bottom rim to the point on the top rim diametrically opposite to it. Taking $\pi = 22/7$, the length of the shortest path (in cm, round off to two decimal places) is _____

Q13 NAT

A solid metal cone of base radius 6 cm and height 8 cm is melted and recast into solid spheres of radius 2 cm each. The number of spheres obtained is _____

Q14 NAT

A fair die is thrown three times. The probability of getting at least one six (round off to three decimal places) is _____

Q15 MCQ

Argument: “Our plant should replace all its old motors with high-efficiency motors, because the new motors use 8 % less electricity.” Which one of the following, if true, most weakens this argument?

(A) Electricity prices are expected to rise next year
(B) High-efficiency motors are now made by several companies
(C) The energy saved over the remaining life of the old motors is worth less than the cost of replacing them
(D) Some of the plant's old motors are more than 20 years old

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Answer Key

Q	Type	Answer
Q1	MCQ	(B) will have finished
Q2	MCQ	(C) extravagant
Q3	MCQ	(A) inconclusive
Q4	NAT	1320
Q5	NAT	30
Q6	MCQ	(D) 35 years
Q7	MCQ	(A) 7.21 km, south-east
Q8	NAT	169
Q9	MCQ	(B) Only I follows
Q10	NAT	20
Q11	MCQ	(C) Figure (C)
Q12	NAT	37.20 (accept 37.00 to 37.40)
Q13	NAT	9
Q14	NAT	0.421 (accept 0.418 to 0.424)
Q15	MCQ	(C) The energy saved over the remaining life of the old motors is worth less than the cost of replacing them

Detailed Solutions

Q1 **Answer: (B) will have finished**

Concept: An action that will be complete before a stated point in the future takes the future perfect tense (will have + past participle).

'By the time the inspector arrives tomorrow' fixes a future point; the test will be over before it: **will have finished (B)**.

'had finished' is past perfect (before a past point); 'will finish' does not show completion before that point.

Exam tip: 'By the time' + present tense in the clause → future perfect in the main clause; 'By the time' + past tense → past perfect.

Q2 **Answer: (C) extravagant**

Concept: Frugal means economical, sparing in the use of money or resources.

Its opposite is wasteful or **extravagant (C)**. 'Thrifty' and 'careful' are near-synonyms, a common trap in antonym questions.

Exam tip: In antonym questions first eliminate the options that mean the same as the given word.

Q3 **Answer: (A) inconclusive**

Concept: The clause after the colon explains the blank: results that settle nothing either way.

Such results are **inconclusive (A)**. 'Conclusive' is the opposite; 'exhaustive' (thorough) and 'impressive' do not fit the explanation.

Exam tip: A colon or 'that is' after the blank usually defines the missing word; let it decide.

Q4 **Answer: 1320**

Concept: SP = CP × (1 − loss %/100); find the cost price first, then apply the required gain.

CP = 1140/0.95 = ₹1200

Required SP = 1200 × 1.10 = **₹1320**

Exam tip: Percentages of profit and loss are always on the cost price, not on the old selling price (1140 × 1.15 = 1311 is wrong).

Q5 **Answer: 30**

Concept: Add the three pair rates: each person is counted twice.

2(A + B + C) = 1/12 + 1/15 + 1/20 = (5 + 4 + 3)/60 = 1/5, so A + B + C = 1/10 per day

A = 1/10 − (B + C) = 1/10 − 1/15 = 1/30 per day, so A alone takes **30 days**

Exam tip: Work with rates (fractions of the job per day), never with the days directly.

Q6 **Answer: (D) 35 years**

Concept: Let the present ages be 7x and 2x; both increase by the same 10 years.

(7x + 10)/(2x + 10) = 9/4 → 28x + 40 = 18x + 90 → x = 5

Father = 7 × 5 = **35 years (D)** (son 10; after 10 years 45 : 20 = 9 : 4 ✓).

Exam tip: Always check the answer in the second ratio; it catches arithmetic slips.

Q7 **Answer: (A) 7.21 km, south-east**

Concept: Track the position on x (east) and y (north) axes, turning with the direction faced.

North 4 → (0, 4); right = east 3 → (3, 4); right = south 8 → (3, −4); left of south = east 3 → (6, −4)

Distance = $\sqrt{(6^2 + 4^2)} = \sqrt{52}$ = **7.21 km, towards the south-east (A)**

Exam tip: 'Right' and 'left' depend on the direction being faced at that moment; write the facing direction after each turn.

Q8 **Answer: 169**

Concept: Look at what kind of numbers the terms are before looking at differences.

4, 9, 25, 49, 121 = 2^2, 3^2, 5^2, 7^2, 11^2: squares of consecutive prime numbers

Next = 13^2 = **169**

Exam tip: If the differences (5, 16, 24, 72) show no pattern, test squares, cubes, primes and factorials.

Q9 **Answer: (B) Only I follows**

Concept: 'Some A are B' + 'No B is C' always gives 'Some A are not C'.

The valves that are pipes cannot be pumps, so some valves are not pumps: I follows.

Nothing links the other valves to pumps; they may or may not be pumps, so II does not follow: **Only I (B)**.

Exam tip: Draw the Venn diagram in the way that tries to break the conclusion; if it can be broken, it does not follow.

Q10 **Answer: 20**

Concept: Percentage excess = (P − Q)/Q × 100, on the base of the quantity being compared with.

Total P = 40 + 46 + 52 + 48 + 60 + 66 = 312; total Q = 36 + 42 + 40 + 50 + 44 + 48 = 260

(312 − 260)/260 × 100 = 52/260 × 100 = **20 %**

Exam tip: 'P exceeds Q by' uses Q as the base; 'Q is less than P by' uses P (16.67 % here).

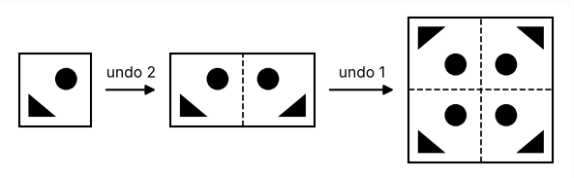
Q11 **Answer: (C) Figure (C)**

Concept: Unfold in the reverse order of folding; each unfolding reflects the holes in that fold line.

Undo fold 2: the holes are reflected in the vertical centre line (the triangles turn to face each other). Undo fold 1: everything is reflected in the horizontal centre line.

The result has the holes in mirror symmetry about both centre lines: **(C)**. (A) copies without reflecting, (B) rotates the holes, and (D) undoes only one fold.

Exam tip: n folds give 2^n copies of each hole; check the count first, then the mirror orientation of a non-symmetric hole.



Q12 **Answer: 37.20 (accept 37.00 to 37.40)**

Concept: Unroll the curved surface into a rectangle; the shortest path becomes a straight line.

Horizontal distance = half the circumference = πr = 22/7 × 7 = 22 cm; vertical distance = 30 cm

Path = $\sqrt{(22^2 + 30^2)} = \sqrt{1384}$ = **37.20 cm**

Exam tip: The same unfolding trick works for a cube or a box: open the faces the path crosses into one plane.

Q13 **Answer: 9**

Concept: Volume is conserved on recasting.

Cone = (1/3)π × 6^2 × 8 = 96π cm^3; sphere = (4/3)π × 2^3 = 32π/3 cm^3

Number = 96π/(32π/3) = **9**

Exam tip: Keep π as a symbol; it cancels and avoids rounding errors.

Q14 **Answer: 0.421 (accept 0.418 to 0.424)**

Concept: 'At least one' = 1 − P(none).

P(no six in three throws) = $(5/6)^3$ = 125/216

P(at least one six) = 1 − 125/216 = 91/216 = **0.421**

Exam tip: Adding 1/6 + 1/6 + 1/6 = 0.5 is wrong because it counts outcomes with two or three sixes more than once.

Q15 **Answer: (C) The energy saved over the remaining life of the old motors is worth less than the cost of replacing them**

Concept: To weaken an argument, attack the unstated assumption linking the evidence to the conclusion.

The argument assumes that saving 8 % of electricity justifies the replacement. If the savings are worth less than the cost, the conclusion fails: **(C)**.

(A) and (D) actually support replacement; (B) is irrelevant.

Exam tip: In GA critical-reasoning questions, find the gap between 'because …' and 'therefore …'; the weakener widens it.

What next?

Day 68 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

Dig Career Thrust app

Telegram: daily PDFs

Next, Day 69/90: EM & SOM Revision. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.